

CUDA Kernel Engineering

Notes from the CudaKernels project — seven kernels, measured and explained

Hardware: NVIDIA GTX 1060 Max-Q (Pascal, sm_61)

Kernels: matmul · dot product · transpose · GEMV · scan · 1D convolution · 2D convolution

Compiled July 2026 — every number was measured on this machine, and every kernel was written by
hand

Contents

Part I — The machine

1. The card you are programming
2. The execution model: grids, blocks, warps
3. The memory hierarchy
4. Measuring honestly: rooflines and ratios

Part II — The kernels

5. Matrix multiplication — reuse and tiling
6. Dot product — reductions and atomics

7. Transpose — coalescing and bank conflicts
8. GEMV — access patterns, and a negative result
9. Scan — dependencies, trees, and barriers
10. 1D convolution — constant memory and halos
11. 2D convolution — the ladder of walls

Part III — The playbook

12. Cross-cutting principles
13. Results and glossary
14. What comes next

1 The card you are programming

Everything in these notes happened on one specific graphics card: a GTX 1060 Max-Q, the mobile version of NVIDIA's mid-range Pascal card from 2016. This matters more than it might seem. Performance work is always a negotiation with a particular machine's limits, and every decision in the later chapters — how big to make a tile, whether a cache will catch some reuse, whether an optimization is even worth attempting — traces back to one of the numbers in this table. It is worth pausing to understand each row before reading on.

Property	Value	Why it matters
GPU	GTX 1060 Max-Q (GP106, Pascal)	compute capability 6.1 (<code>sm_61</code>)
SMs (streaming multiprocessors)	10	the independent "cores" that run blocks
fp32 peak	~3.9 TFLOP/s	the ceiling for arithmetic
DRAM peak bandwidth	~192 GB/s	the ceiling for streaming data
L2 cache (chip-wide)	1.5 MB	shared by all SMs
Shared memory per SM	96 KB	the pool that blocks lease slices of
Max threads per SM	2048	64 warps; the basis for occupancy
Warp size	32	the true unit of execution

The two numbers to memorize are the two ceilings. The card can perform at most about 3.9 trillion single-precision floating-point operations per second, and it can move at most about 192 billion bytes per second between its memory chips and the processor. Every kernel is limited by one of these two, and much of the craft in this project was figuring out *which one* a given kernel was pressed against — because effort spent on the wrong ceiling is wasted.

One more thing colors every measurement in these notes: this is a laptop card. Its clock speed rises and falls with temperature, so the same kernel can run ten percent faster or slower from one run to the next, and for some kernels the swing across a cold machine and a hot one reached fifty percent. Two habits protect against being fooled by this. First, every benchmark takes several samples and reports the median. Second — and more important — whenever two versions of a kernel are compared, they are run back-to-back inside the same program execution, so that both see the same thermal conditions and their *ratio* is trustworthy even when the absolute times are not.

2 The execution model: grids, blocks, warps

When you launch a CUDA kernel, you describe the work as a **grid** of **blocks**, where each block contains up to 1024 **threads**. You choose both shapes at launch time with the triple-angle-bracket syntax: `kernel<<<grid, block>>>(...)`. The hardware then hands out whole blocks to the SMs, and each block stays on its SM until it finishes.

That is the programmer's view. The hardware's view contains one more level, and it is the most important idea in this entire document. Inside the SM, threads are not executed one at a time. They are bundled into groups of 32 called **warps**, and a warp executes as a single unit: one instruction is fetched, and all 32 threads — called **lanes** when we think of them this way — carry it out at the same moment, in lockstep. You write a program for one thread, but the machine only ever runs it 32 threads at a time.

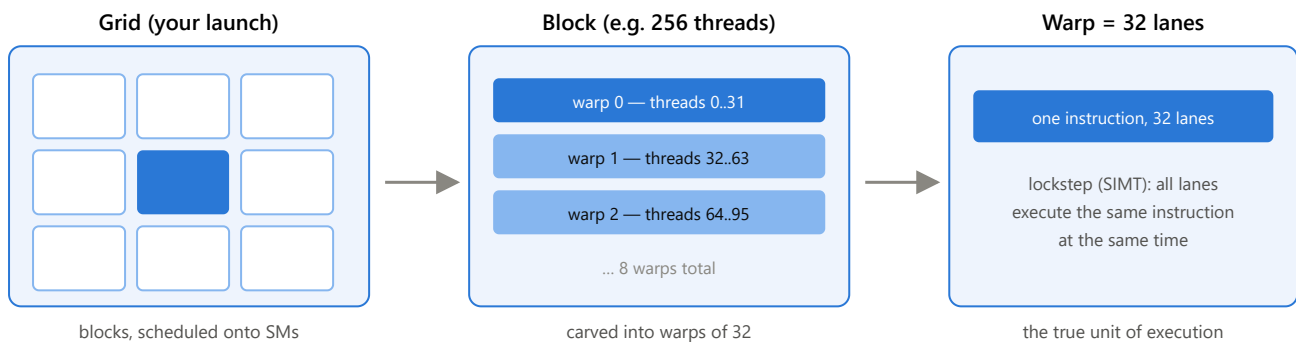


Figure 2.1 — The three levels: the grid you launch, the blocks the hardware schedules, and the warps of 32 lanes that actually execute. Nearly every performance effect in these notes — coalescing, divergence, bank conflicts, shuffles — happens at the warp level.

2.1 Which threads share a warp depends on the block's shape

Threads are packed into warps in order of their linear index within the block, and the x coordinate varies fastest. In a one-dimensional block of 256 threads, warp 0 is simply threads 0 through 31. But in a two-dimensional 16×16 block, the first 32 threads are the first *two rows* of the block — so a warp covers two adjacent rows of 16 threads each. Reshape the same 256 threads into a 32×8 block, and a warp becomes exactly one row of 32. This sounds like a cosmetic detail. It is not: in chapter 11, this exact difference — whether a warp lies along one row or straddles two — decides whether every shared-memory read in the kernel runs at full speed or half speed.

2.2 What lockstep means for your code

Lockstep execution has three consequences that echo through every kernel in Part II.

The first concerns **branching**. If the 32 lanes of a warp reach an `if` and disagree about the condition, the warp cannot split in two — there is only one instruction stream. Instead the hardware runs one side of the branch with the disagreeing lanes switched off, then the other side with the roles reversed. This is called **divergence**, and its cost is the sum of both paths. Often, though, the compiler avoids branching entirely using **predication**: every lane executes the instruction, and a small per-lane flag decides whether the result is kept or thrown away. Sections 2.4 through 2.7 open this machinery up in detail — predication in particular overturns the usual intuition about what an `if` costs.

The second concerns **memory**. Because a warp issues its 32 loads together, the hardware has the opportunity to merge them into just a few memory transactions — but only if the addresses cooperate. This is called **coalescing** and is the subject of the next chapter.

The third is a gift: lanes of a warp can pass values to each other directly, register to register, using **shuffle** instructions — no memory involved at all. Chapter 6 builds a complete reduction out of them.

2.3 Latency hiding: why the GPU wants thousands of threads

A single read from the card's DRAM takes several hundred clock cycles to arrive. A CPU hides such delays with large caches and clever speculation. A GPU does something simpler: it keeps many warps resident on each SM, and the moment one warp stalls waiting for memory, the SM switches to another warp that is ready. The stalled warp's delay is hidden behind other warps' work.

This only works if there *are* other warps to switch to. The number of warps resident on an SM — its **occupancy**, out of a maximum of 64 on this card — is therefore a resource in its own right. Anything that reduces it, such as a block asking for a large slice of shared memory or using many registers per thread, shrinks the SM's supply of "other work" and leaves memory delays exposed. Chapter 8 measures exactly this effect going wrong.

2.4 Inside the SM: four small machines

To understand what the SM actually does with its warps, we need to open it up. An SM's 128 arithmetic cores are not one undifferentiated pool. They are organized into **four partitions** of 32 cores each, and at the front of every partition sits a small unit called a **warp scheduler**. When blocks are placed on the SM, their warps are divided among the four schedulers — with the SM fully loaded, each scheduler owns sixteen of the 64 resident warps — and each warp belongs to its scheduler for life.

Notice how deliberately the widths line up: one scheduler issues one instruction for one warp of 32 lanes, and its partition contains exactly 32 cores to carry that instruction out. A partition is, in effect, "everything one warp-instruction needs for one cycle." The SM is best pictured not as one big processor but as four small ones side by side, each juggling its own sixteen warps.

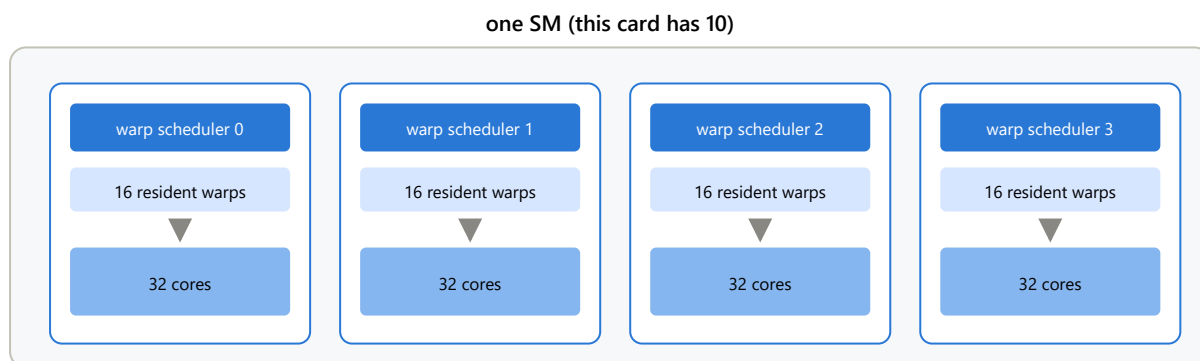


Figure 2.2 — Inside one SM: four partitions, each with its own scheduler, its own sixteen warps, and its own 32 cores. Every warp's registers live permanently in the partition's register file for as long as the warp is resident — which is why, as §2.6 explains, switching between warps moves no data and costs no time.

2.5 Issuing: the SM's serial currency

All the work an SM ever does begins the same way. On each clock cycle, each scheduler picks one warp — from its own sixteen — whose next instruction is able to start, and starts it: hands it to the arithmetic units, or the memory units, whichever the instruction needs. That act of starting an instruction is called **issuing** it, and each scheduler can issue for only about one warp per cycle. The execution units downstream are wide and run in parallel, but nothing reaches them except through a scheduler, one instruction at a time. A helpful picture is a ticket window: every instruction a warp will ever execute must pass through the window, the window serves one instruction per cycle, and the queue behind it never gets to skip a turn.

Those per-cycle opportunities are called **issue slots**, and they are the SM's serial currency — the thing every instruction must spend exactly one of. That makes the card's total issue capacity a simple multiplication. Each scheduler starts one warp-instruction per cycle, which does the work of 32 threads; four schedulers per SM makes 128 thread-instructions per cycle; ten SMs at roughly 1.4 GHz gives about **1.8 trillion thread-instructions per second** for the whole card. And that number puts a hard floor under every kernel's runtime, independent of anything memory does:

$$\text{minimum time} \geq \text{total thread-instructions} \div 1.8 \text{ trillion per second}$$

This floor is not a curiosity — it decided a real optimization. In chapter 11, the first 2D convolution kernel spends about ten instructions per tap, of which only one is the multiply-add doing mathematics. Multiply it out: 16.7 million pixels × 81 taps × 10 instructions is about 13.6 billion instructions, and dividing by the issue capacity gives 7.5 milliseconds — a floor the kernel could not go below *even if memory were infinitely fast*. The kernel measured 10.6 ms, so it was pressed against this wall, not the memory wall, and the optimizations that followed were all one campaign: make the program shorter, so fewer trips through the ticket window are needed. (A footnote for completeness: schedulers can occasionally start two instructions from the same warp in one cycle when the pair needs different execution units — so the true capacity is slightly higher — and an issued instruction then takes further cycles to *complete* its journey through the units. Neither detail changes the picture: issue slots are scarce and serial, and every instruction costs one.)

2.6 A cycle in the life of a scheduler

Now let us watch the scheduler make its choices, because the mechanism is the whole secret of GPU latency hiding. Each warp carries its own **program counter** — a marker pointing at the next instruction it has not yet started — and its own registers, held permanently in the partition's register file. Every cycle, the scheduler asks one question: *which of my sixteen warps has a next instruction that can start right now?* An instruction can start when the values it needs exist — nothing it depends on is still in flight — and the unit it needs has room. A warp that fails this test is **stalled**: not paused mid-action, simply ineligible this cycle.

Follow a concrete story, drawn in Figure 2.3. Warp 1's first five instructions are each ready when their turn comes, so the scheduler issues them on five consecutive cycles. Instruction 5 is a load from DRAM, and instruction 6 needs the loaded value — which will not arrive for some 400 cycles. Warp 1 is now stalled. The scheduler does not wait with it: on the very next cycle it finds another eligible warp — warp 2, say, which had been stalled and whose awaited value arrived a moment ago — and issues warp 2's next instruction, from exactly where warp 2's program counter points. Nothing is skipped and nothing repeated; stalling never loses a warp's place, because the program counter *is* the place. For the next four hundred cycles the scheduler weaves among its other warps, and warp 1's enormous memory delay disappears behind their work. When the value finally lands, warp 1 becomes eligible again, and some cycle soon after, instruction 6 issues as if nothing had happened.

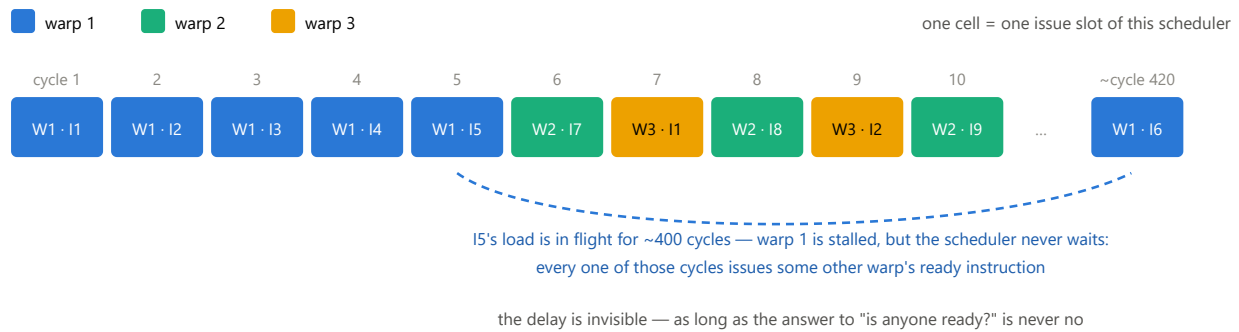


Figure 2.3 — One scheduler's issue slots over time. Warp 1 issues five instructions, then stalls on a memory load; the scheduler immediately interleaves warps 2 and 3, each resuming from its own program counter, and warp 1's instruction 6 issues hundreds of cycles later when the loaded value arrives. Occupancy is what keeps this weave possible: the more resident warps, the rarer the cycle where nobody is ready and the slot is wasted.

Two properties of this dance are easy to miss and worth stating plainly.

First, **switching warps costs nothing** — and this is the deep difference from a CPU. When a CPU switches threads it must save one thread's registers and load another's, a matter of thousands of cycles, so it switches rarely and reluctantly. On the GPU, all sixteen warps' registers sit permanently side by side in the partition's register file, and every warp keeps its own program counter. "Switching" from warp 1 to warp 2 is nothing more than the scheduler selecting a different warp number this cycle. No state moves anywhere. That is why the scheduler can alternate between warps on back-to-back cycles, at a granularity no operating system could dream of.

Second, **the choosing itself consumes no cycles**. It is natural to imagine the scheduler pausing to scan its sixteen warps — but that intuition comes from software, where scheduling is a program that takes time. Here scheduling is *wiring*. A piece of bookkeeping hardware (traditionally called a **scoreboard**) maintains a running ready-or-not flag for every warp, updated by events as they happen: when warp 1 issues its load, warp 1's flag drops; four hundred cycles later, the arriving data itself flips the flag back. Nothing ever has to go looking. And picking one warp from sixteen ready-flags is a small combinational circuit that settles within a fraction of the cycle, concurrently with the current instruction being dispatched — the ticket window does not close while deciding whom to serve next. The scheduler's cost is paid in silicon and power, not in time. This is also why real scheduling policies are simple (issue from the same warp until it stalls, then move to the oldest ready one): a cleverer policy would be a bigger, slower circuit, and a 0.7-nanosecond cycle will not wait for it.

The one thing that *can* go wrong is an empty queue. If all sixteen flags say "not ready" — typically because too few warps are resident and they all hit their memory stalls together — the cycle's issue slot is simply lost. This is occupancy's whole story told from the scheduler's chair, and it is exactly what the GEMV experiment of chapter 8 measures: shared memory demands cut occupancy to 37.5%, each scheduler was left juggling about six warps instead of sixteen, the "is anyone ready?" question started coming up empty, and the kernel slowed down even though no single instruction got slower.

2.7 What an `if` really costs: divergence and predication

Section 2.2 introduced the two ways a warp copes with an `if` — divergence and predication — in a sentence each. Now that we know what issue slots are, we can do the accounting properly, because the two mechanisms are best understood as two ways of paying the same bill.

Start from the constraint that creates the bill. A warp has one instruction stream. Whatever its 32 lanes need done, all of it must pass through that single stream — so if lane 3 needs path A and lane 7 needs path B, the stream must contain *the union of both paths*, and each lane sits idle through the part that is not its own. That is the

essence of divergence, and notice that it is a statement about lanes disagreeing, not about branches. No encoding can make disagreement free; the union is the union. Branching and predication are just two layouts for it:

A **branch** jumps over work no active lane needs. When the warp is unanimous, it executes only the side it took. When lanes disagree, the hardware runs each side in turn with the other side's lanes masked off, then reconverges.

Predication never jumps. Both sides sit in the stream unconditionally; the compare writes a per-lane true/false flag, and every following instruction carries the flag, taking effect only in lanes where it says yes. In the compiled 2D convolution this looks like `@!P3 LDG ...` followed by `@!P3 FFMA ...` — load and multiply-add that happen, per lane, only where flag P3 permits.

To reason about predication precisely, separate two questions that sound alike. *Does the warp spend time on a predicated instruction?* Always — it is next in the stream, so it consumes an issue slot (§2.5) whether the flags enable 32 lanes or none. The scheduler cannot peek at flags and skip ahead; deciding to skip *is* branching, the very thing predication exists to avoid. *Does the instruction take effect in a given lane?* Only if that lane's flag says so. A masked lane's destination register is untouched, and — crucially — a masked lane's **memory access never happens**: a predicated-off load sends no request to the memory system at all. This is a correctness guarantee, not just a performance fact. In chapter 11's kernel D, edge threads compute addresses that point *outside the image* before the bounds check decides anything; the reason that is safe is that the masked load never touches memory, so the bad address is only ever arithmetic in a register. In short: **predication always spends the warp's time, but only conditionally spends each lane's effects.**

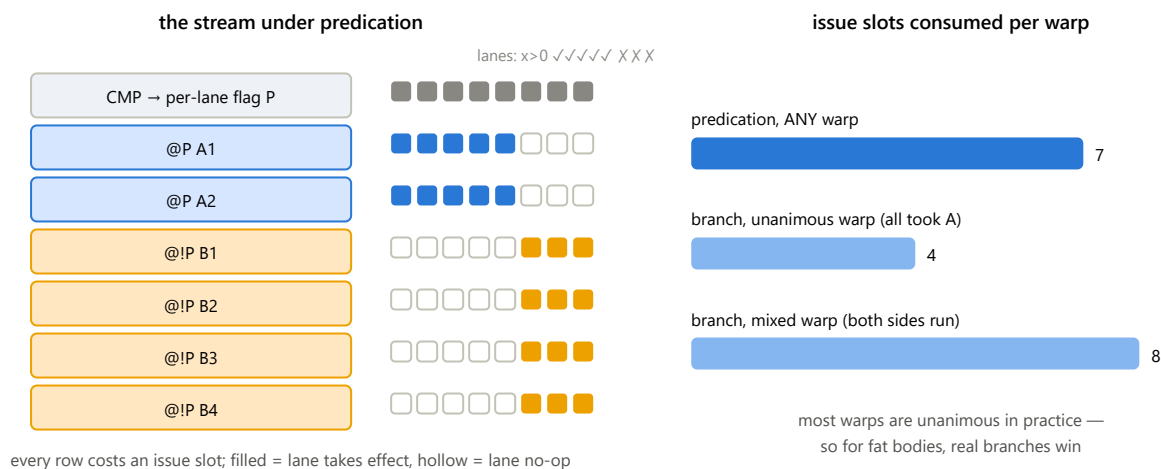


Figure 2.4 — An `if` with a 2-instruction A-side and a 4-instruction B-side. Left: under predication the stream contains everything, every row costs an issue slot for every warp, and the flags only control which lanes keep results. Right: the slot count per warp. Predication charges the union to everyone, always; a branch charges the union only to warps whose lanes actually disagree. When bodies are small the difference is negligible and predication's simplicity wins; when a body is fat, branching wins because most warps are unanimous.

The figure's arithmetic explains when each mechanism wins, and it is exactly the trade the compiler weighs for every `if` it compiles. Predication is the better deal when the guarded work is small — a load and a multiply-add, say — because "just issue it everywhere and let the flags sort it out" costs less than the jumping-and-reconverging machinery. This is why the bounds check in the 2D convolution kernel compiled to predicated instructions, and why an edge warp there costs exactly as much as an interior warp. But predication charges every warp for both sides, forever. Give the `else` a hundred instructions, and predicated code makes the thousands of warps that would unanimously skip it pay for all hundred anyway — while a branch lets them leap over it for the price of a compare and a jump. Real kernels' warps are unanimous far more often than not, so for fat bodies the

branch wins by a wide margin, and the compiled 2D convolution contains both mechanisms side by side: predication for the small guarded loads, genuine branches around the larger blocks. (Predication has a second, quieter cost: both sides' intermediate values must be computed before the flags choose, so both sides' values need registers at once — and registers, like occupancy, are a budget.)

Finally, there is work predication cannot express at all: control flow whose *shape* depends on data. The canonical example is a loop whose iteration count differs per lane — `while (error > tolerance)`, where each lane's error converges at its own pace. No assignment of per-instruction flags can make lane 5 run the body three times while lane 20 runs it nine hundred times; the number of instructions in the stream would itself have to differ per lane. All the machine can do is keep the whole warp looping until the slowest lane finishes, with finished lanes masked off and idle — divergence in its pure, irreducible form. When you meet it, the remedy is never an encoding trick; it is redistributing the work so the lanes of a warp want similar amounts, which is precisely the challenge the sparse matrix-vector kernel on the roadmap exists to teach.

WHAT TO REMEMBER

The warp scheduler is the machine's beating heart: four per SM, each making a free, per-cycle choice among its sixteen warps, with switching costing nothing because every warp's registers and program counter stay put. Issue slots are the serial currency — about 1.8 trillion thread-instructions per second on this card — and dividing a kernel's instruction count by that capacity gives a floor no memory optimization can breach. Divergence's true cost is the union of the work lanes disagree about; predication and branching are just two ways to lay that union out, spending time unconditionally but effects per-lane. And some divergence — lane-varying loop counts — no mechanism removes; only reshaping the work does.

3 The memory hierarchy

A GPU's memory is a hierarchy of levels, each smaller and faster than the one below it. What makes GPU programming different from CPU programming is that the two most useful levels are under *your* control, not the hardware's. The figure shows the hierarchy with this card's numbers.

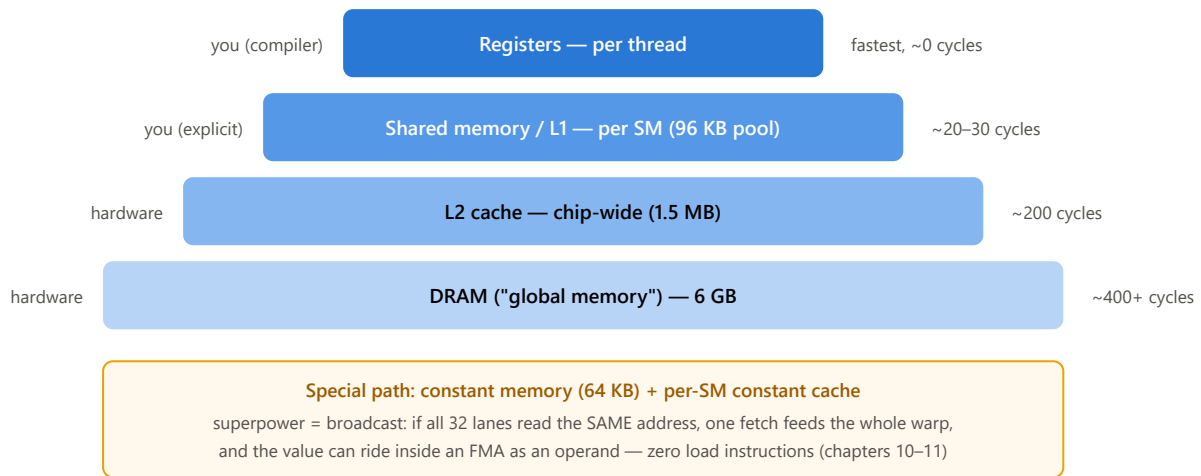


Figure 3.1 — The hierarchy with this card's numbers. The labels on the left say who manages each level. The two levels you control directly — registers and shared memory — are where every optimization in Part II lives.

3.1 Coalescing: the first rule of global memory

The card's DRAM is not read one number at a time. It is read in chunks: 32-byte sectors, grouped into 128-byte cache lines. Now recall from chapter 2 that a warp issues its 32 loads together. If those 32 loads ask for 32 *consecutive* floats — 128 contiguous bytes — the hardware satisfies the entire warp with a single 128-byte transaction. Every byte fetched is a byte the program wanted. This is a coalesced access, and it is the best case.

Now suppose instead that each lane reads an address one full matrix row away from its neighbor. The 32 requests now touch 32 different cache lines, so the hardware must make 32 separate transactions, and each 128-byte line it fetches contains only 4 bytes the program actually asked for. Same useful data, up to thirty-two times the traffic. Chapter 8 measures this exact difference on real code: 43 GB/s against 151 GB/s, with identical arithmetic.

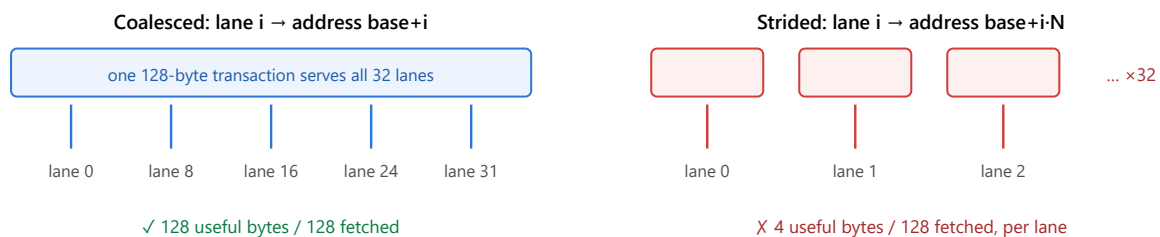


Figure 3.2 — Coalescing. Both sides deliver the same 32 floats of useful data; the strided pattern generates up to 32× the memory traffic. The lesson: design the mapping from threads to data so that consecutive lanes touch consecutive addresses.

3.2 Shared memory: a leased scratchpad, not a cache

Each SM owns 96 KB of very fast on-chip memory. When a block is placed on an SM, it receives an exclusive slice of this pool, sized to whatever the kernel requested, and it keeps that slice until the block finishes. Arrays declared `__shared__` live in the slice, and all threads of the block — but only that block — can read and write them. It helps

to picture the SM as a landlord with 96 KB of floor space, leasing a fixed amount to each tenant block for the duration of its stay.

Three consequences follow from this arrangement, and each one cost us a lesson somewhere in Part II. First, shared memory is **a purchase, not a free win**: every kilobyte a block requests is floor space no other block can lease, so a greedy request directly reduces how many blocks fit on the SM — and with them, the occupancy that hides memory latency. In chapter 8, a 16 KB request cut occupancy from 100% to 37.5% and made the kernel *slower*. Second, a new block's slice **starts as garbage** — whatever the previous tenant left behind. Any slot your compute code will read must first be written by your load code; forgetting this produces silent wrong answers, not crashes. Third, the slice is **private to the block**. Two blocks cannot share data through shared memory even if they happen to sit on the same SM; when blocks must cooperate, the data has to travel through global memory between kernel launches, which is exactly how the multi-block scan of chapter 9 is structured.

3.3 Banks: how shared memory is wired

Shared memory is divided into **32 banks**. A bank's word is 4 bytes wide, and word i lives in bank $i \bmod 32$ — so 32 consecutive floats occupy all 32 banks, and the pattern repeats every 128 bytes. The reason for this design is that all 32 banks can be read in the same cycle: a warp gets its 32 values at once *provided each lane hits a different bank*. (There is one exception: if several lanes want the very same word, the hardware broadcasts it for free.)

The trouble begins when two lanes want *different* words from the *same* bank. The bank can only serve one of them per cycle, so the access replays — twice as slow for a 2-way conflict, up to 32 times slower if a whole warp lands on one bank. Bank conflicts appear three times in these notes — in transpose, in scan, and in 2D convolution — and, instructively, they demanded three different fixes.

3.4 A quirk of this card that shaped chapter 11

THE SM_61 QUIRK

On this card, an ordinary load from global memory never uses the small L1 cache inside the SM — it always travels to the chip-wide L2, about 200 cycles away. The fast path through the per-SM cache is reserved for a special **read-only load instruction**, and the compiler may only emit it when it can prove that no thread writes the loaded memory at any point during the kernel. Writing `const` on the pointer is not proof, because the same memory might be written through a different pointer. The annotation that does constitute proof is `const float* __restrict__` — the programmer's signed promise that the pointers do not overlap. In chapter 11, adding two such annotations was worth a 1.4× speedup on its own.

4 Measuring honestly: rooflines and ratios

4.1 The roofline: which ceiling are you under?

Every kernel does two things: arithmetic, and data movement. Divide one by the other — useful floating-point operations per byte that the kernel *must* move — and you get its **arithmetic intensity**. This single number tells you which of the card's two ceilings applies. If the intensity is low, the memory system runs out of capacity long before the arithmetic units do, and the kernel is **memory-bound**: its best possible speed is $\text{intensity} \times 192 \text{ GB/s}$. If the intensity is high, memory keeps up easily and the kernel is **compute-bound**, capped at 3900 GFLOP/s. The crossover point — where the two ceilings meet — is at $3900 \div 192 \approx 20.3$ operations per byte on this card, and it is called the **ridge**.

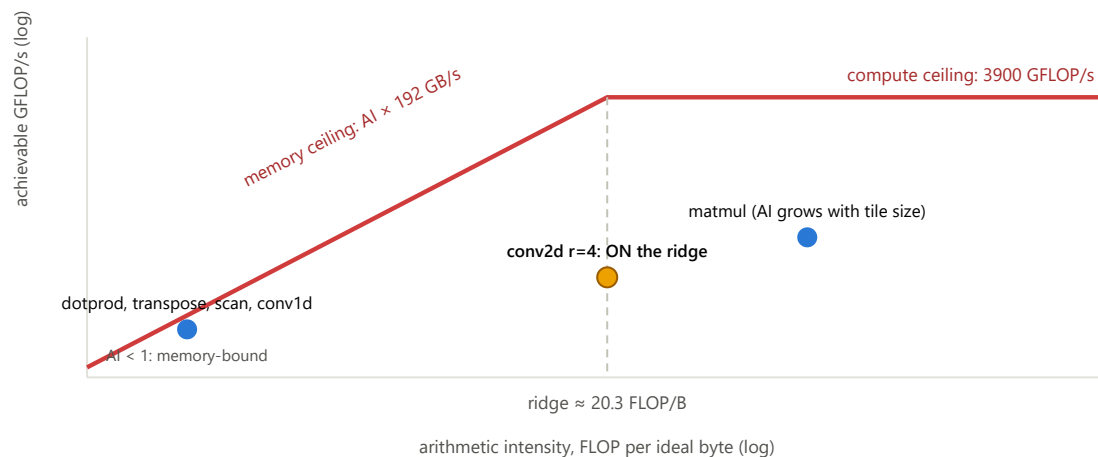


Figure 4.1 — The roofline model for this card, drawn schematically. To the left of the ridge only bandwidth matters; to the right, only arithmetic throughput. Most kernels in this project live far to the left. The 2D convolution at radius 4 lands exactly on the ridge, which is why chapter 11 reports both a GB/s and a GFLOP/s number and watches whichever is further from its peak.

4.2 Effective bandwidth: measure against the ideal

All of the GB/s figures in these notes are **effective bandwidth**: the bytes the kernel *logically must* move — typically, read each input once and write each output once — divided by the time it took. Notice what this deliberately leaves out. If a kernel re-reads the same input many times, or fetches 128 bytes to use 4, that extra traffic does not count as useful work; it just makes the effective number smaller. This turns effective bandwidth into a score against perfection. On this card, a perfectly-behaved streaming kernel sustains about 110–140 GB/s in practice, so a kernel that reaches that range is finished: it wastes nothing, and the remaining gap to the 192 GB/s spec sheet belongs to the memory chips, not to the code.

4.3 The measurement discipline

Five habits kept the numbers in this project honest, and they are worth stating as rules. Benchmarks run a warm-up pass and then report the **median** of several timed samples, because a laptop's clock speed wanders. Competing kernel versions are compared **within a single run**, so the comparison cancels out whatever thermal state the machine is in. Correctness always gates performance: every kernel is checked against a slow, obviously-correct CPU reference before it is ever timed. The output buffer is **poisoned** with NaN bytes before each kernel runs, so a kernel that silently writes nothing fails loudly instead of passing on stale data. And test inputs are

chosen to break lucky symmetries — odd image sizes that exercise every edge, and an asymmetric convolution mask, so that a transposed index cannot accidentally pass.

One more instrument deserves mention because it recurs. The usual profiler was unavailable on this machine (Windows blocks its hardware counters without administrator settings), so the project learned to work without it: controlled parameter sweeps that decompose a kernel's time into physically meaningful parts (chapter 10), and reading the compiler's actual output — the GPU assembly — to count instructions directly (chapter 11).

5 Matrix multiplication — reuse and tiling

Matrix multiplication is where the project began, and it introduces the single most important idea in GPU performance: find the **reuse** hidden in a computation, and then decide which level of the memory hierarchy will capture it. We wrote three versions of the same kernel. The naive one reached about 179 GFLOP/s. Adding a shared-memory tile raised that to about 430. Making each thread compute four outputs instead of one raised it again, to just over 1000 GFLOP/s — about a quarter of the card's arithmetic peak. Each jump came from moving reused data one level closer to the arithmetic units.

5.1 Why the naive kernel is slow

The natural first kernel assigns one thread to each element of the output matrix C. That thread walks across a row of A and down a column of B, multiplying and accumulating. The arithmetic is fine; the problem is what it does to memory. Every element of A is read not once but by an entire row of threads, and every element of B by an entire column — the same numbers are fetched from global memory over and over. For each pair of 4-byte values loaded, the thread performs one multiply-add, which works out to roughly a quarter of a floating-point operation per byte. Look back at the roofline in Figure 4.1: that intensity is far to the left of the ridge, deep in memory-bound territory. And so the textbook compute-bound problem runs, in its naive form, as a memory-bound kernel at five percent of the card's arithmetic peak.

5.2 Tiling: buy the reuse with shared memory

The cure is to stop asking global memory for the same values repeatedly. Partition C into 16×16 tiles and give each tile to one block. The block then marches along the shared dimension of A and B in 16-wide steps. At each step, its 256 threads cooperatively load one 16×16 patch of A and one of B into shared memory — each thread carries exactly one element of each, and because neighboring threads load neighboring elements, the loads coalesce. Then every thread computes its 16 multiply-adds entirely out of the two shared patches, and the block steps forward:

```
for (int t = 0; t < numTiles; t++) {
    Ads[ty][tx] = /* guarded load of one element of A's tile */;
    Bds[ty][tx] = /* guarded load of one element of B's tile */;
    __syncthreads();                // wait until both tiles are complete
    for (int k = 0; k < TILE_WIDTH; k++)
        value += Ads[ty][k] * Bds[k][tx];
    __syncthreads();                // wait until everyone is done reading
}
```

Count what happened to the traffic. A 16×16 patch is loaded from global memory once — 256 loads — and then each of its values is read sixteen times by threads working out of shared memory. The kernel asks global memory for one sixteenth of what the naive version asked for, and the reuse now lives in memory that is roughly ten times closer. That is the entire trick, and it more than doubled the speed.

Notice the two `__syncthreads()` barriers. The first ensures no thread starts computing from a tile that some other thread has not finished filling. The second ensures no thread races ahead and overwrites the tile while a slower thread is still reading it. This load–barrier–compute–barrier rhythm is the canonical shape of every tiled kernel in this project; it will return in chapters 7, 10 and 11.

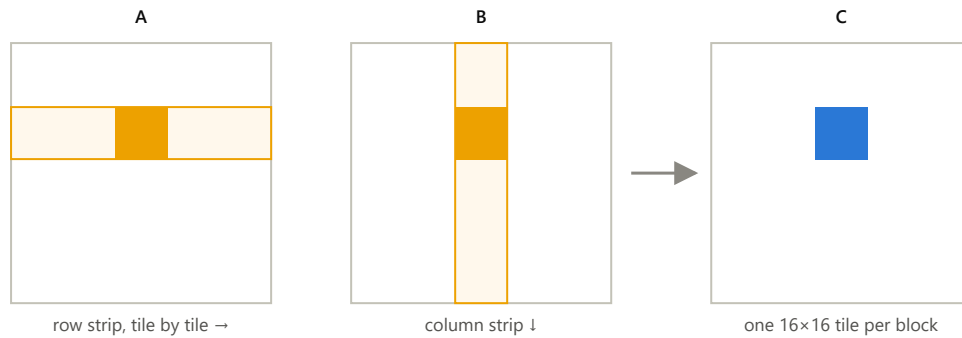


Figure 5.1 — Tiling. To produce the blue tile of C, the block walks the orange strips of A and B one 16×16 patch at a time. Each patch is fetched from global memory once and then read sixteen times from shared memory. A larger tile would buy even more reuse per byte fetched — tile size is how matmul climbs the roofline toward the compute ceiling.

5.3 Register tiling: reuse within a single thread

Shared memory is fast, but a register is faster still — and the tiled kernel above reads shared memory twice for every multiply-add it performs. The third version improves that ratio by making each thread responsible for a 2×2 square of outputs rather than a single element. In the inner loop, the thread reads two values from the A patch and two from the B patch — four shared-memory reads — and combines them in all four ways:

```
for (int k = 0; k < TILE_WIDTH; k++) {
    c00 += Ads[ty][k] * Bds[k][tx];
    c01 += Ads[ty][k] * Bds[k][tx + TILE_WIDTH];
    c10 += Ads[ty + TILE_WIDTH][k] * Bds[k][tx];
    c11 += Ads[ty + TILE_WIDTH][k] * Bds[k][tx + TILE_WIDTH];
}
```

Four reads now feed four multiply-adds instead of two feeding one — twice the arithmetic per shared-memory access — and the four accumulators live in registers, the fastest storage the machine has. This change alone took the kernel from 430 to over 1000 GFLOP/s. The idea generalizes and will reappear at the end of chapter 11 as the known next step for convolution: when shared memory becomes the bottleneck, give each thread more outputs so that each value loaded is used more times.

WHAT TO REMEMBER

The three versions of matmul are one lesson told three times: the computation has enormous reuse, and performance is decided by where that reuse is captured. Left to the caches, it crawls. Captured in shared memory, it doubles. Captured partly in registers, it doubles again. There is still headroom — the kernel reaches only 26% of the card's arithmetic peak, and the known next technique (double-buffering: loading patch $t+1$ while computing on patch t) is on the project roadmap.

6 Dot product — reductions and atomics

The dot product of two vectors of 16.7 million floats is, arithmetically, almost nothing: one multiply and one add per pair of elements. All the difficulty is in combining sixteen million partial results into a single number without the combining itself becoming the bottleneck. The finished kernel streams the two input vectors at about 130 GB/s — roughly 70% of the spec-sheet bandwidth, which is the practical ceiling — and the entire reduction machinery is invisible in the timing. This chapter is about that machinery, because the project reuses it everywhere.

6.1 Grid-stride loading

Rather than launching one thread per element, the kernel launches a fixed-size grid (capped at 1024 blocks) and lets every thread process many elements, striding by the total number of threads:

```
for (int i = idx; i < N; i += gridDim.x * blockDim.x)
    sum += A[i] * B[i];
```

This little pattern earns its keep three ways. At every step of the loop, consecutive lanes read consecutive addresses, so the loads coalesce perfectly. The same launch handles any input size without recomputing grid dimensions. And when the loop finishes, each thread holds exactly one partial sum in a register — the raw material for the reduction.

6.2 Reducing: warp, block, grid

The reduction proceeds in three stages, one for each level of organization.

Within a warp, the lanes combine their 32 partials using shuffle instructions. `__shfl_down_sync(mask, val, offset)` lets lane i read the register of lane $i + \text{offset}$ directly — no shared memory, no barrier. Adding with offsets of 16, 8, 4, 2 and finally 1 folds all 32 values into lane 0 in five steps:

```
for (int offset = 16; offset > 0; offset >>= 1)
    val += __shfl_down_sync(0xffffffff, val, offset);
```

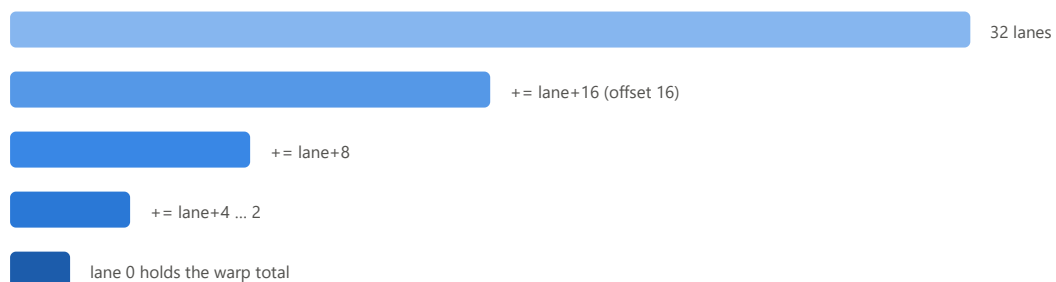


Figure 6.1 — The warp shuffle reduction. Each row is one step; the live values halve each time until lane 0 holds the sum of all 32. Five register-to-register steps, no memory and no barriers. This helper function was written once for the dot product and reused verbatim in GEMV.

Within a block, each warp's lane 0 deposits its warp total into a tiny shared array of at most 32 entries. After one barrier, the first warp picks those entries up and shuffle-reduces them the same way, producing a single block total.

Across the grid, thread 0 of each block adds its block total into the final answer with a single `atomicAdd`. An atomic addition is safe under contention but serializes when many threads target the same address — which is

exactly why the grid was capped: at most 1024 atomic additions for 16.7 million inputs is negligible. One practical detail: the accumulator must be zeroed before launch, and the launcher does this itself. An accumulator-style kernel has a different contract with its caller than an overwrite-style kernel, and it is worth noticing which one you are writing.

WHAT TO REMEMBER — INCLUDING TWO NEGATIVE RESULTS

The finished kernel is memory-bound, and that is the success criterion: the reduction adds nothing visible to the cost of streaming the data. Two "obvious" optimizations were then tried and honestly measured.

Vectorized `float4` loads made the kernel about 5% *slower* — coalesced scalar loads already saturate the memory transactions on this card, and the wider loads reduced the number of loads in flight. Unrolling the loop four ways changed nothing outside noise. Optimizations that sound obviously good frequently are not; measure, and keep the simple version when it wins.

7 Transpose — coalescing and bank conflicts

Transposing a matrix computes nothing at all — every microsecond of its runtime is data movement, which makes it the cleanest laboratory for memory behavior in the whole project. The three versions tell the story in numbers: the naive kernel moved data at about 56 GB/s, staging the transpose through shared memory raised that to 82, and a one-character fix to the shared array's declaration raised it to 139 — about 72% of the card's peak, close to the practical ceiling.

7.1 A structural problem

The transpose writes $B[j][i] = A[i][j]$. Suppose we assign threads so that a warp reads 32 consecutive elements of a row of A — a perfectly coalesced read. Those 32 values belong to 32 different *rows* of B , so the warp's 32 writes land a full row apart from each other: the worst-case strided pattern of Figure 3.2. Flip the assignment and the writes become clean but the reads fall apart. As long as threads carry values straight from A to B , one side must suffer the $32\times$ penalty.

7.2 Staging the swap through shared memory

The escape is to change the data's orientation somewhere the change costs nothing. Each block loads a 32×32 tile of A into shared memory row by row — coalesced reads — and waits at a barrier. Then it writes the tile out to B , but reading the *shared* array transposed (`tile[tx][ty]` instead of `tile[ty][tx]`) and swapping the roles of the block coordinates. The result is that the global *writes* also sweep along rows of B — coalesced as well. Both global-memory sides of the kernel now enjoy their preferred access order, and the actual transposition happens inside shared memory, where addresses are cheap. Almost.

7.3 The bank conflict, and the one-character fix

Reading `tile[tx][ty]` means each warp reads a *column* of the shared tile. With a row length of 32 floats, walking down a column means visiting addresses 0, 32, 64, 96, ... — every one of them congruent to the same value mod 32, which by the rule of §3.3 means every one of them lives in the **same bank**. All 32 lanes queue up at one bank while the other 31 banks sit idle: the read replays 32 times, and this is precisely why the staged version stalled at 82 GB/s.

The fix is famous and satisfying. Declare the tile with one extra, never-used column:

```
__shared__ float tile[32][33]; // 33 ≠ multiple of 32 → columns spread across banks
```

The row length is now 33, so successive elements of a column sit 33 words apart — and because 33 shares no factor with 32, those addresses fall into banks 0, 1, 2, 3, ... in turn. The column has been smeared diagonally across all 32 banks, and the conflict is gone. One character of source code, 32 wasted words per tile, and a $1.7\times$ speedup.

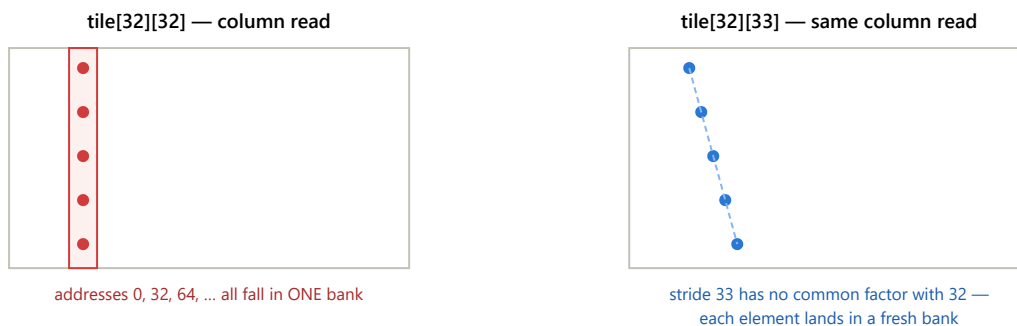


Figure 7.1 — The padding trick. On the left, a column of the tile lives entirely in one bank and the warp's read serializes 32 ways. On the right, the extra padding column makes the row stride 33, coprime with the 32 banks, so the same logical column drifts diagonally through all of them. The cost is 32 unused words per tile; the gain was 1.7×.

WHAT TO REMEMBER

When the two sides of a kernel want incompatible memory orders, stage the data through shared memory and give each side the order it wants. Then check the shared-memory accesses themselves: a bank conflict is invisible in the source code and enormous in effect, and the way to find it is to reason about each warp's addresses modulo 32. The coprime-stride pad returns in scan (as `i + (i>>5)`), and chapter 11 introduces a bank conflict this trick *cannot* fix — knowing which kind you are facing matters.

8 GEMV — access patterns, and a negative result

GEMV multiplies a 4096×4096 matrix by a vector. It is a modest computation, but it produced two of the most valuable lessons in the project: the largest single speedup — 43 to 151 GB/s from changing nothing but the assignment of threads to data — and the most instructive failure, an optimization that made the kernel slower for two reasons that compound.

8.1 Thread-per-row against warp-per-row

The natural mapping gives each thread one row of the matrix: thread t computes $y[t]$ by walking row t and accumulating. To see why this is a disaster, stop thinking about one thread and look at what a *warp* does at each step. At column i , lane 0 reads $A[0][i]$, lane 1 reads $A[1][i]$, lane 2 reads $A[2][i]$ — addresses an entire row apart, 16 KB from each other. That is the strided pattern of Figure 3.2 in its purest form, and the kernel achieved about 43 GB/s.

The repair inverts the mapping: give each row to a **warp** instead of a thread. Now lane l reads columns l , $l+32$, $l+64$ and so on — at every step, the warp's 32 lanes touch 32 consecutive addresses, one coalesced transaction. When the row is exhausted, each lane holds a partial sum, and the shuffle reduction from chapter 6 collapses the 32 partials so that lane 0 can write the answer:

```
for (int i = lane; i < N; i += 32)
    sum += A[wid * N + i] * x[i];
sum = warpReduceSum(sum);
if (lane == 0) y[wid] = sum;
```

The arithmetic is identical to before. Only *who reads what* changed, and throughput went from 43 to 151 GB/s — the cost of uncoalesced access, isolated and measured.

8.2 The shared-x experiment: a negative result worth keeping

An observation invites one more optimization: every row multiplies against the same vector x , which is 16 KB. Why not have each block copy x into shared memory once, and let its warps read the copy? The reasoning sounds airtight. It was implemented carefully — and the kernel got fifteen to twenty percent *slower*. Two effects, stacking:

First, **the L2 cache already had it**. The vector is re-read by all 4096 rows, so after the first few blocks touch it, all 16 KB of x sits comfortably in the chip-wide 1.5 MB L2 cache. The "slow global reads" the staging loop was built to eliminate were already being served from L2. The staging saved almost no DRAM traffic and added real work: 4096 extra loads, 4096 shared-memory stores, and a barrier in every block.

Second, **occupancy collapsed**. Each block now requested 16 KB from its SM's 96 KB pool, so at most six blocks fit per SM where sixteen fit before. Resident warps fell from 64 to 24 — from 100% to 37.5% occupancy — and with so few other warps available, the SM could no longer hide the latency of the (unavoidable) loads of the matrix A itself.

THE RULE THIS FAILURE BOUGHT

Shared memory is a purchase. The price is occupancy plus staging work plus a barrier; the return is the number of reuses multiplied by the saving per access *compared to wherever the data would otherwise live*. That final clause is the trap: if the data already fits in L2, the saving per access is nearly nothing and the purchase is a loss. Matmul tiles and convolution halos pay off handsomely; a small read-only vector does not. Before staging anything into shared memory, ask: where would the caches put this anyway?

9 Scan — dependencies, trees, and barriers

Scan, or prefix sum, computes running totals: `out[i]` is the sum of everything up to and including element `i`. Every kernel before this chapter was "embarrassingly parallel" — each output depended only on a handful of nearby inputs, so threads never needed to wait for one another. Scan is different in kind: the last output depends on *every* input before it. Parallelizing it forces two new kinds of thinking — restructuring arithmetic into trees, and coordinating blocks that cannot talk to each other. It also produced the project's most humbling measurement, in which an asymptotically superior algorithm bought seven percent and a footnote-level fix bought seventy.

9.1 Two ways to scan a block

Within one block, where threads can share memory and synchronize, there are two classic algorithms.

The **Hillis-Steele** scan is the simple one. In the first round, every element adds the element one place to its left. In the second round, every element adds the element *two* places left; then four, then eight, and so on. After $\log_2 N$ rounds, every element has accumulated everything to its left. The subtlety is a race: an element must not be overwritten before its right neighbor has read it, so each round reads into a register, waits at a barrier, writes, and waits again. The cost of the simplicity is extra work — N additions per round, $\log_2 N$ rounds, so $O(N \log N)$ additions where a sequential scan does N .

The **Blelloch** scan repairs the work count with a beautiful two-phase tree. The **up-sweep** phase builds sums of pairs, then pairs of pairs, and so on up an implicit binary tree laid over the array — after it finishes, the last element holds the total of the whole block. Then the last element is cleared to zero, and the **down-sweep** walks back toward the leaves, at each node swapping the left child's value with the running total and adding it in. What emerges is an *exclusive* scan (each position holds the sum of everything strictly before it), and the total work is $O(N)$ — asymptotically as good as sequential. Hold on to that fact; the punchline of this chapter is what it turned out to be worth.

9.2 Crossing block boundaries: the three-pass structure

A million elements need a thousand blocks, and blocks cannot see each other's shared memory. The only cheap way all blocks can agree that some stage of work is finished is the boundary between one kernel launch and the next. So the full scan runs as three launches. In **pass one**, every block independently scans its own 1024-element chunk and writes one extra number: the chunk's total. In **pass two**, a single block scans the array of chunk totals, turning each into an offset — the sum of all chunks to its left. In **pass three**, every element simply adds its chunk's offset, and the independent chunk-scans become one global scan. Global memory serves as the mailbox between passes.

9.3 The upset: measuring where the time actually goes

Profiling the three passes revealed that pass one consumed about 80% of the runtime — and that it ran at 26 GB/s while pass three, which touches *the same amount of data* but has no barriers, ran at 117 GB/s on the same machine. The gap between those two numbers is not memory; it is synchronization. Pass one was **barrier-bound**: roughly twenty `__syncthreads()` per block, each one forcing every warp to wait for the slowest.

That diagnosis makes a prediction. Swapping Hillis-Steele for Blelloch in pass one cuts the arithmetic from $O(N \log N)$ to $O(N)$ — but it does *not* reduce the number of barriers, which is about twenty in both. And indeed, the swap bought a factor of just **1.07**. The algorithms textbook promised a large improvement; the wall clock, which bills for barriers rather than additions, disagreed.

The real villain was hiding in Blelloch's memory addresses. The tree walk touches elements at strides of 1, 2, 4, ... and at a stride of 32, all the active lanes land in the *same shared-memory bank* — the 32-way pile-up of \$3.3, occurring level after level. The fix is the transpose pad of chapter 7 translated to one dimension: insert one padding word after every 32 real words, by remapping every index i to $i + (i \gg 5)$. That single remapping bought a factor of **1.7** on pass one — ten times more than the better algorithm did.

Change to pass 1	What theory promised	What the clock said
work-efficient algorithm ($O(N \log N) \rightarrow O(N)$)	large speedup	1.07×
pad shared indices: $i \rightarrow i + (i \gg 5)$	a footnote	1.7×

WHAT TO REMEMBER

Kernel launches are the only cheap grid-wide synchronization, so multi-block algorithms are designed as passes with global memory as the mailbox. And Big-O analysis counts arithmetic, but on a GPU the expensive things are barriers and bank conflicts — so profile before trusting asymptotics, and compare candidate fixes by same-run ratios. There is known headroom left in this kernel (a better global-load layout, and fusing pass three into pass one), recorded on the roadmap.

10 1D convolution — constant memory and halos

A convolution slides a small window of weights — the **mask** — along a signal. Each output is a weighted sum of its neighborhood: $\text{out}[i] = \sum \text{in}[i+j] \cdot \text{mask}[j+r]$ for j from $-r$ to r , with positions past the ends of the signal counting as zero. With 16.7 million elements and a radius of 4 (nine weights, or "taps"), the four versions of this kernel climbed from 29 GB/s to about 110 GB/s — and 110 GB/s is this card's real streaming ceiling, which means the final version is finished in the strongest sense: it moves each byte once, at full speed, with all the arithmetic hidden behind the memory traffic.

Two features make convolution a teaching kernel. Every input element is reused by $2r+1 = 9$ neighboring outputs, and every thread reads the same tiny mask. Those are two different kinds of reuse, and they are served by two different tools.

10.1 The mask belongs in constant memory

Watch a warp execute the tap loop. At iteration j , every lane computes `mask[j + r]` — an index that depends on the loop counter but *not* on the thread. All 32 lanes read the same address at the same moment, every iteration. CUDA has a memory space built precisely for this pattern: **constant memory**, a 64 KB read-only region with a small dedicated cache in each SM whose special ability is **broadcast** — when all lanes of a warp ask for the same address, one cache access feeds all 32. Better still, the compiler can fold a constant-memory value directly into a multiply-add instruction as an operand, so reading the weight stops costing an instruction at all.

Moving the mask from an ordinary pointer into constant memory made the kernel 2.8× faster. It is worth being precise about why, because the obvious explanation is wrong. The mask is 36 bytes; it was already sitting in the L2 cache either way. The win came from the instruction stream: the change removed nine load instructions and nine 200-cycle L2 round-trips from every output's nine-tap dependency chain. Proximity was never the issue — broadcast and operand-folding were. (The converse pattern is worth knowing too: if lanes index constant memory *differently* from one another, the reads serialize, and constant memory becomes slower than the alternatives.)

10.2 The input belongs in a tile — with a halo

Now for the input's reuse. A block computing outputs B through $B+255$ needs inputs $B-4$ through $B+259$: its own range, plus r extra elements on each side. Those extra elements are called the **halo** (or ghost cells) — inputs the block needs but whose outputs belong to the neighboring blocks. The tiled kernel has the block cooperatively copy its entire input range, halo included, into shared memory before computing:

```
for (int s = threadIdx.x; s < tileSize; s += blockDim.x)
    tile[s] = (inRange(g)) ? in[g] : 0.0f;      // g = blockStart - r + s
__syncthreads();
for (int j = -r; j <= r; j++)
    sum += c_mask1d[j + r] * tile[r + threadIdx.x + j];
```

Notice the elegant side effect of the load loop: positions outside the signal are written as literal zeros *into the tile*. The boundary handling is baked into the data once, at load time — and the compute loop, which runs nine times per thread, contains no bounds checks at all. This idea, "bake the padding in, keep the hot loop branch-free," becomes the centerpiece of chapter 11.

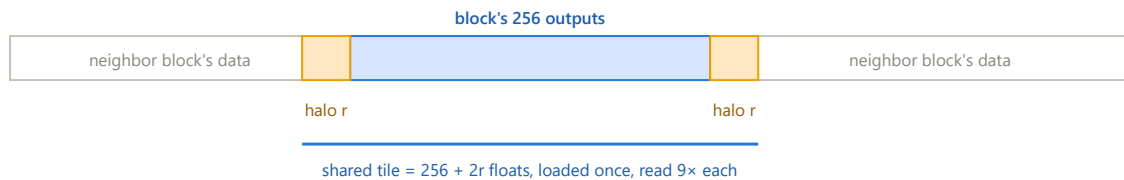


Figure 10.1 — The halo. The orange cells are inputs this block needs even though their outputs belong to the neighboring blocks; each block loads its own copy of the shared border. Out-of-range positions become zeros in the tile, so the compute loop never checks bounds.

10.3 A sweep instead of a profiler

With the profiler's hardware counters blocked on this machine, the project built its own instrument: run every variant at radii 1, 2, 4 and 8, and watch how time grows with the number of taps. The results organize themselves beautifully. Every variant shares the same fixed cost — about 1.2 milliseconds, which is exactly the time to stream 128 MB through memory at the card's real speed — and each variant adds a per-tap price on top. That price is a direct readout of *where each tap's operands live*: about 0.45 ms per tap for the naive kernel (two L2 round-trips — input and mask), about 0.10 for the constant-mask kernel (one L2 trip — the mask is now free), and about 0.04 for the tiled kernel (shared memory).

The same sweep exposes a break-even point that the single-radius benchmark would have hidden. Tiling has a fixed overhead — the tile load and its barrier — so it only pays once enough taps use the tile. At radius 1 and 2 the constant-mask kernel actually wins; the tiled kernel pulls ahead only from radius 4. This is chapter 8's rule wearing a different coat: shared memory is a purchase, and at two taps of reuse the purchase does not cover its own cost.

10.4 Telling the compiler the radius: `template <int R>`

One inefficiency remains in the tiled kernel. Because the radius r arrives at runtime, the compiler must emit a genuine loop for the taps: increment a counter, compare, branch — bookkeeping on every iteration — and it cannot begin all nine shared-memory reads at once. But in practice the radius is one of a handful of small values. Making it a C++ template parameter, `template <int R>`, produces a separate compiled kernel for each radius of interest (a `switch` in the launcher picks the right one, with the generic kernel as a fallback for unusual radii). Within each specialized kernel, R is a compile-time constant, so `#pragma unroll` can flatten the tap loop into straight-line code: no counter, no branch, and all nine loads issued together so their latencies overlap. This final version gained another 1.4× and reached the streaming floor — its time barely changes between three taps and nine, because the memory traffic, not the arithmetic, is all that remains.

WHAT TO REMEMBER

When every lane of a warp reads the same address, that read belongs in constant memory — the win is broadcast plus operand-folding, not proximity. When a block's outputs need overlapping inputs, load the block's range plus a halo into shared memory and write the boundary zeros into the tile, so the hot loop stays branch-free. A parameter sweep can decompose runtime into a floor plus a per-item price and reveal where operands live — a homemade profiler that is sometimes better than the real one. And any value the compiler can know at compile time is optimization fuel; templates are the mechanism for feeding it.

11 2D convolution — the ladder of walls

The 2D convolution slides a 9×9 window of weights over a 4096×4096 image — 81 taps per pixel, radius 4, zero padding at the edges. It is the capstone of the project for two reasons. First, its arithmetic intensity (162 operations per 8 ideal bytes, about 20 per byte) lands exactly on the roofline ridge of Figure 4.1, making it the first kernel where *both* ceilings matter at once. Second, its optimization story had a shape the earlier chapters only hinted at: four optimizations, where **each performance wall was invisible until the previous one fell**. Three predictions in a row missed before one landed — and each miss identified a wall nobody had priced in. The versions ran at 15.0, 10.6, 8.4, and finally 3.9 milliseconds.

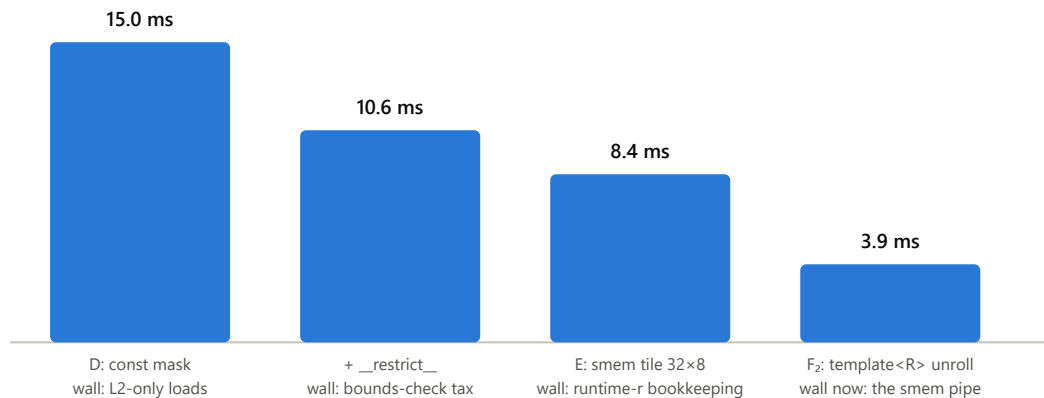


Figure 11.1 — The ladder. Beneath each bar is the wall that capped that version — that is, the thing the next step had to remove. The total improvement is 3.9 $\times$, and the final kernel spends about three quarters of its time feeding the shared-memory pipeline, which is the honest definition of "the tile is doing its job."

11.1 What changes when convolution goes 2D

The first version, kernel D, is simply the 1D constant-mask kernel in two dimensions: a 2D grid of 16×16 blocks, one thread per pixel, two nested tap loops, a bounds check for the zero padding, and the mask in constant memory. What changes is the geometry of the memory traffic. The reuse explodes — each input pixel now feeds 81 outputs instead of 9. And the reuse splits into two very different kinds. Along a row, a warp's nine horizontal taps fall within about two cache lines, so they are nearly free re-reads. Down a column, however, vertically adjacent pixels sit a full image row apart in memory — 16 KB — so the nine vertical taps touch nine different cache lines, and the threads that could reuse them belong to *different blocks*, scheduled at unpredictable times on unpredictable SMs. Whether any cache catches that vertical reuse is suddenly a serious question.

11.2 Step one: `__restrict__` (15.0 → 10.6 ms)

Kernel D came in three times slower than even a pessimistic prediction, and the diagnosis was the quirk of §3.4: on this card, every one of the 81 taps was traveling to the L2 cache, because ordinary loads bypass the small cache inside the SM. Estimating the traffic confirmed it — about 5.4 GB of L2 reads over 15 ms is roughly 370 GB/s, which is about what the L2 can deliver. The kernel was saturating the L2 while the DRAM sat nearly idle at 9 GB/s: a bottleneck one level *above* the one everyone usually watches.

The fix is a promise. Writing `const float* __restrict__ in, float* __restrict__ out` tells the compiler that these pointers do not overlap — that nothing written through `out` can ever change the bytes read through `in`. The compiler needs this proof to be about the whole kernel, not one thread, because 16.7 million threads run concurrently: even though each thread's store happens after its own loads, another thread's store could race with this thread's loads if the buffers overlapped. With the promise in hand, the compiler switched every input read to

the read-only load instruction (visible in the compiled assembly as `LDG.E.CI`), the horizontal re-reads started hitting the cache inside the SM, and 4.4 milliseconds fell away. The promise costs nothing here, because calling this kernel with overlapping buffers was already a bug — in-place convolution would race regardless.

11.3 An aside: what does that bounds check actually cost?

A natural worry about kernel D's `if` — the zero-padding bounds check executed for all 81 taps — is branch divergence. The measured truth is more interesting. First, divergence requires lanes *of the same warp* to disagree, and in a 4096^2 image only the blocks touching an edge (about 1.6% of them) ever produce disagreement; for everyone else the condition is uniformly true. Second, the compiler did not emit a branch at all: it used predication (§2.7), compiling the guarded load and multiply-add with a per-lane flag, so an edge warp and an interior warp execute the identical instruction sequence in identical time.

So the check causes no divergence to speak of. What it does cost is **instructions**. Counting the compiled loop revealed about ten instructions per tap, of which exactly one was the multiply-add doing the mathematics; three or four of the others were computing "yes, this tap is in bounds" — an answer that is a foregone conclusion for 98.4% of warps. That reframes the question an `if` should raise in a kernel. Ask first whether lanes within a warp disagree often (here, almost never), and then ask what evaluating the condition costs even when everyone agrees (here, nearly half the kernel's instructions). The second question is usually where the money is, and it cannot be fixed by making the condition cheaper — only by removing the condition altogether.

11.4 Step two: the halo-ring tile (10.6 → 8.4 ms)

Removing the condition altogether is exactly what the 1D playbook prescribes: stage the block's input neighborhood in shared memory with the zeros baked in. In two dimensions the halo becomes a **ring**: a block computing a 32×8 patch of pixels needs that patch plus a border of width 4 on all four sides — a 40×16 region, 640 values, about 2.6 KB of shared memory. (Why 32×8 rather than 16×16 is the subject of §11.5.) Global reads drop from 81 per pixel to 2.5, since the tile is loaded once and neighboring blocks re-read only the thin border they share.

Two mapping problems are genuinely new in 2D, and both are worth internalizing.

The first is the **load**. The block has 256 threads but must fill 640 tile slots, so there is no one-slot-per-thread assignment. The pattern that solves it is called flatten-and-stride: number the slots 0 through 639 in row-major order and let thread t fill slots t , $t+256$, and $t+512$, exactly like the 1D tile loop. Each flat slot number is converted back to a row and column by dividing by the **row width** — slot s is tile row $s \div 40$ and column $s \bmod 40$ — and each tile position maps to the global pixel that sits r rows up and r columns left of the block's corner. If that pixel lies outside the image, the thread stores 0.0f, and the padding is baked in. Because consecutive slots are consecutive columns, the global reads coalesce.

The second is the **compute indexing**, and it contains both a pleasant surprise and the kernel's classic bug. The surprise: because the tile begins r up-and-left of the output patch, thread (ty, tx) 's 9×9 window starts *exactly* at tile position (ty, tx) — no $+r$ offsets anywhere in the loop. The bug: the tile's rows are 40 wide, not 32, so every row index must be multiplied by the tile's own stride, `side = 40`. Using the block width instead produces output that is smooth, plausible-looking, and wrong everywhere — which is precisely why the test suite includes odd-sized images (333×479) whose failures are unmistakable.

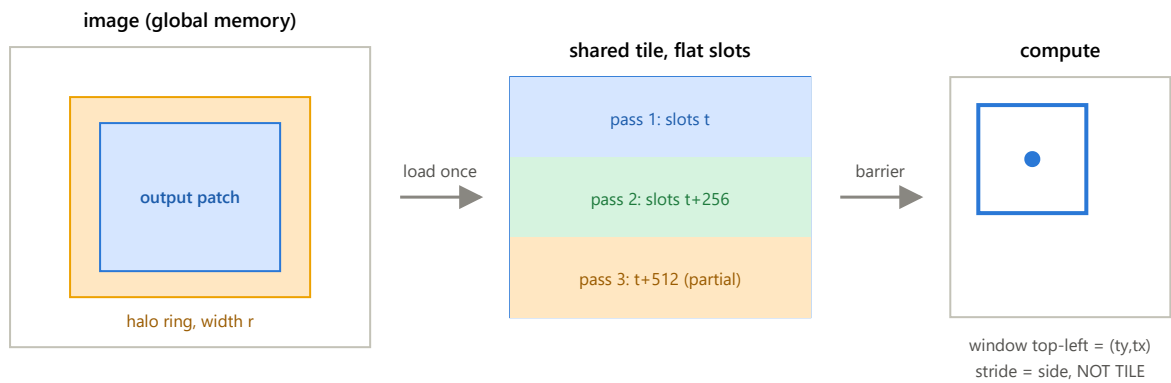


Figure 11.2 — Kernel E. The block loads its output patch plus the halo ring into shared memory in strided passes, waits at the barrier, and computes without a single bounds check. One discipline is critical: a thread whose pixel falls outside the image must still help load and must still reach the barrier — only its final write is skipped. An early return before a barrier is a hang. The interactive version of this figure is docs/conv2d_halo_tiling_viz.html.

11.5 An aside: bank conflicts you cannot pad away

The tile was originally planned for 16×16 blocks, and working through the bank arithmetic of §3.3 for that shape reveals a trap. In a 16×16 block, a warp is two rows of 16 threads (§2.1). During any single tap, the first half-warp reads 16 consecutive tile words starting at some address, and the second half-warp reads 16 consecutive words starting one tile-row later — 24 words away, for a 16-wide tile with radius 4. Take everything modulo 32 and the two segments overlap: eight lanes of the second half land on the same banks as eight lanes of the first, at different addresses. Every single shared-memory read in the kernel would be a 2-way bank conflict, running the shared pipeline at half speed.

The transpose trick — pad the stride — does not work here. Shifting the row stride from 24 to 25 just rotates which eight lanes collide. The collision comes not from the stride but from the *shape of the warp itself*: two 16-word segments can never occupy 32 banks unless they are exactly 16 apart modulo 32, and no reasonable tile width arranges that. The fix is to change the shape: a **32×8 block** has the same 256 threads, but each warp is now a single row of 32 (§2.1 again), reading 32 consecutive words — 32 distinct banks, conflict-free by construction, for any stride.

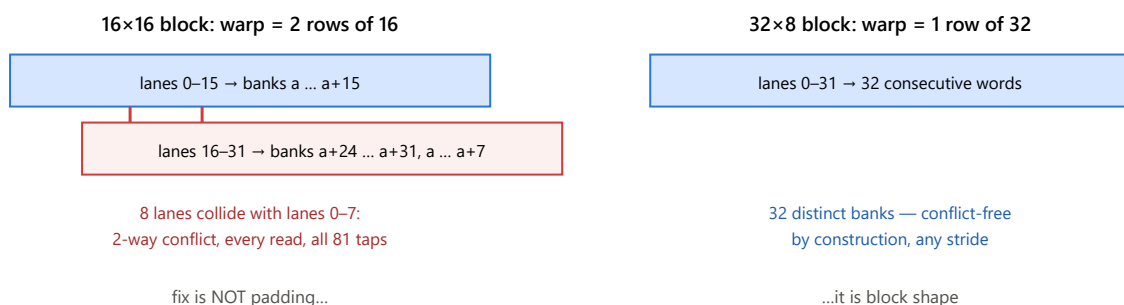


Figure 11.3 — The same 256 threads in two shapes. Transpose's bank conflict came from a strided access pattern, which padding fixes. This one comes from the warp straddling two short rows, which only reshaping fixes. Diagnose which species you have before reaching for the familiar cure.

Now the honest twist, and it may be the most valuable lesson in the chapter. Making this fix produced **no measurable improvement at all** — the reshaped kernel ran in the same 8.4 ms as the square one. The conflict was real, but its cost lived in the shared-memory pipeline, and the kernel was not yet limited by the shared-memory pipeline; it was limited by instruction issue, a wall that sat above it. A bottleneck fixed below the

waterline does not move the boat. The fix was kept anyway, because the next optimization was about to lower the waterline — and one step later it became worth a factor of two.

11.6 Step three: the compile-time radius (8.4 → 3.9 ms)

Why did the tiled kernel stop at 8.4 ms? Reading the compiled assembly answered it. With the radius arriving at runtime, the compiler had actually produced an excellent inner loop that processes sixteen taps at a time — and guarded it with a test that only takes that path when more than twelve taps remain in the current row. Our rows have nine taps. The beautiful fast path was dead code at every radius we care about, and the kernel spent its life in fallback loops, re-deriving addresses between rows, with each tap paying for a separate constant-memory load of its mask weight — roughly five instructions and *two* load-type operations per tap.

The cure is the one chapter 10 already taught: `template <int R>`, a `switch` in the launcher, and `#pragma unroll` on *both* tap loops. With the radius a compile-time constant, all 81 taps flatten into straight-line code, every mask index becomes a constant, and each weight rides inside its multiply-add instruction as a constant operand — the separate load disappears entirely, exactly as in 1D:

```
FFMA R4, R5, c[0x3][0x4], R4    // multiply-add with the mask weight
                                // baked in as an operand – no load instruction
```

Counting the compiled kernel confirmed something rare: per pixel, **exactly 81 shared-memory loads and 81 multiply-adds** — two instructions per tap, zero mask loads, zero loop bookkeeping. The kernel dropped to 3.9 ms, right inside the predicted range, and the arithmetic of what remains is tidy: those shared-memory loads, at the hardware's rate of one warp-wide load per cycle per SM, account for about 3 of the 3.9 milliseconds. The kernel now spends three quarters of its time doing the one thing it cannot avoid — and this is where the 32×8 reshape finally pays, because with the square block shape, every one of those loads would take two cycles instead of one, and the floor would sit near 6 ms rather than 3.

WHAT TO REMEMBER

Performance walls fall one at a time, and each is invisible until the one in front of it falls — so predictions will miss, and each miss is a diagnosis. On this card, `__restrict__` is nearly free speed whenever overlapping buffers would already be a bug. The cost of an `if` is usually the condition's evaluation, not divergence, and the cure is structural (bake padding into data), not cleverer branching. Bank conflicts come in two species — strided access (pad it) and warp shape (reshape it). An optimization can be correct, measure zero, and still be right — if the wall it removes is next in line. And when the profiler is locked, `cuobjdump -sass` plus patient counting is a profiler nobody can block. The known headroom, for a future session: each thread computes several neighboring outputs, whose windows overlap 8/9 — the register-tiling idea from `matmul`, returning in stencil form.

12 Cross-cutting principles

Seven kernels produced a small set of rules that kept proving themselves. Each one below is stated in full and then anchored to the chapter where it was paid for, so that when a rule stops sounding convincing, you can revisit the measurement that made it true.

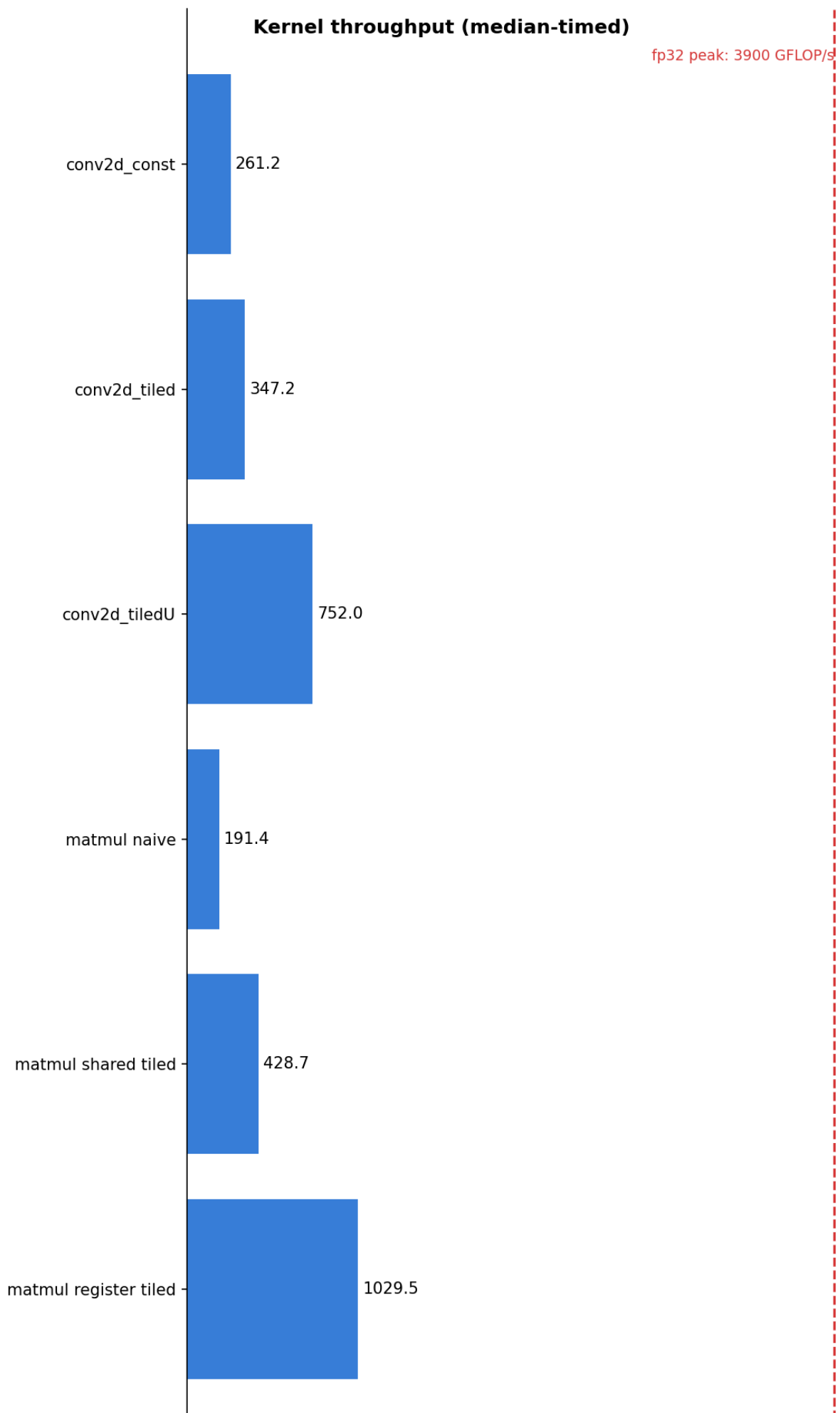
1. **Think in warps, not threads.** The thread is the programming abstraction; the warp is what the machine actually runs. Coalescing, shuffle reductions, divergence, predication, and bank conflicts are all warp-level phenomena — you cannot reason about any of them from a single thread's point of view. (Chapters 3, 6, 7, 11.)
2. **Design the thread-to-data mapping before anything else.** The two largest single improvements in the entire project — GEMV going from 43 to 151 GB/s, and transpose from 56 to 139 — involved no new hardware features at all. They came from changing which thread touches which data, so that warps read consecutive addresses. (Chapters 7, 8.)
3. **Find the reuse, then choose where it lives.** Data that is used once should stream; data that is reused belongs as close to the arithmetic as its reuse justifies: the caches, the constant cache for warp-uniform reads, shared memory for a block's working set, registers for a thread's own accumulations. Matmul climbed this ladder once; convolution climbed it again. (Chapters 5, 10, 11.)
4. **Shared memory is a purchase.** Its price is occupancy, staging work, and a barrier; its return is the number of reuses times the saving per access compared to where the data would otherwise live. The GEMV shared-x experiment is the canonical overpayment — the data was already in L2, and the occupancy cost made the kernel slower. The conv1d radius sweep even located the exact break-even point. (Chapters 8, 10.)
5. **Barriers and bank conflicts outrank Big-O.** The work-efficient scan algorithm bought 7%; removing its bank conflicts bought 70%. Asymptotic analysis counts arithmetic, but the GPU bills for synchronization and serialization. (Chapter 9.)
6. **The cost of an `if` is usually the condition, not the divergence.** Predication means the guarded work runs everywhere anyway; what you pay for, on every warp including the ones that never disagree, is evaluating the condition — nearly half of kernel D's instructions. Remove checks structurally, by baking the boundary into the data. (Chapters 10, 11.)
7. **Give the compiler compile-time facts.** A runtime loop bound blocks unrolling, keeps bookkeeping instructions alive, and forces operands to be loaded rather than folded into instructions. A template parameter plus a dispatch switch turned ten instructions per tap into two. (Chapters 10, 11.)
8. **Walls fall one at a time.** An optimization aimed below the currently binding wall will measure zero — which proves it early, not wrong. Keep it if the roadmap says its wall is next; re-measure after every wall falls. The 32×8 reshape measured zero, then became worth a factor of two one step later. (Chapter 11.)
9. **Trust ratios, not absolutes — and let correctness gate performance.** On thermally drifting hardware, only same-run comparisons mean anything. And a fast wrong kernel is worthless: poisoned outputs, odd sizes, and asymmetric masks caught real bugs that friendlier tests would have blessed. (Chapters 1, 4.)
10. **Negative results are results.** Vectorized loads regressed; unrolling was neutral; shared-x regressed; the better algorithm underwhelmed; a correct fix measured zero. Each is recorded alongside its reason, and together they form a better education than the wins. (Chapters 6, 8, 9, 11.)

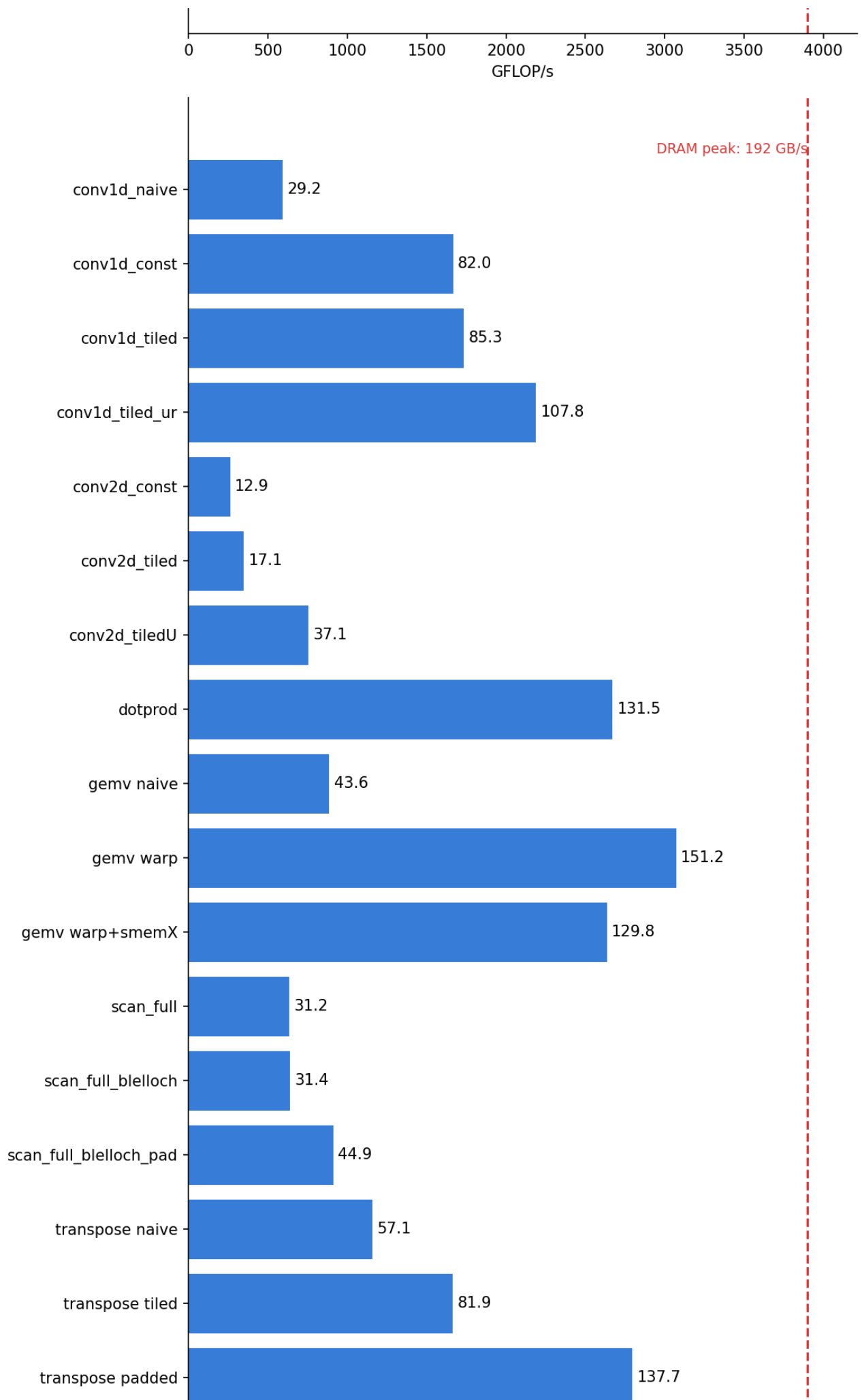
11. **When the profiler is locked, build instruments.** A parameter sweep that separates fixed cost from per-item price; a same-run A/B that isolates one variable; an instruction census read straight from the compiled assembly. None of these can be blocked by a driver setting. (Chapters 4, 10, 11.)

13.1 Every measured result in one place

The table records the final measured numbers for each kernel's ladder of versions, in the order the versions were built. All were taken on the GTX 1060 Max-Q described in chapter 1, using median-of-samples timing.

Kernel	Size	Versions, in order	Measured
matmul	1024-class	naive, shared-tiled, register-tiled 2×2	~179, ~430, ~1044 GFLOP/s
dot product	16.7M	grid-stride + shuffle reduction + atomic	~125–138 GB/s
transpose	2048×4096	naive, tiled, padded [32][33]	~56, ~82, ~139 GB/s
GEMV	4096 ²	naive, warp-per-row, + shared-x (regression)	~43, ~151, ~130 GB/s
scan	1M	3-pass: Hillis-Steele, Blelloch, + padded indexing	pass-1 ratios 1 : 1.07 : ~1.7
conv1d	16.7M, r=4	naive, const-mask, tiled, template-unrolled	~29, ~82, ~87, ~112 GB/s
conv2d	4096 ² , r=4	const-mask, + __restrict__, tiled 32×8, template-unrolled	15.0, 10.6, 8.4, 3.9 ms





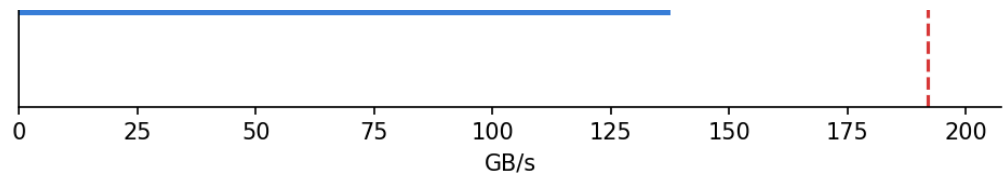


Figure 13.1 — The project scoreboard, `docs/perf.png`. It is regenerated at the end of every session by a script that runs every test binary and refuses to plot anything if a single correctness check fails.

13.2 Glossary

Terms are listed roughly in the order the notes introduce them.

Term	Meaning
warp / lane	a group of 32 threads that execute each instruction together, in lockstep / one thread's position (0–31) within its warp
occupancy	how many warps are resident on an SM, out of a maximum of 64 — the SM's supply of "other work" for hiding memory latency
warp scheduler	the unit at the front of each SM partition (four per SM) that picks one ready warp per cycle and starts its next instruction
issue / issue slot	starting an instruction / one scheduler's per-cycle opportunity to do so — the SM's serial currency, which every instruction costs exactly one of
program counter	a warp's marker for the next instruction it has not yet started; stalling never loses the place, because this is the place
scoreboard	the always-on bookkeeping hardware that keeps a ready-or-not flag per warp, updated by events (a load result arriving flips the flag) rather than by scanning
coalescing	the hardware merging a warp's 32 loads into a few wide transactions, possible when consecutive lanes read consecutive addresses
shared memory	the 96 KB scratchpad inside each SM; blocks lease private slices of it for their lifetime
bank / bank conflict	shared memory is 32 banks, word i in bank $i \bmod 32$; two lanes needing different words of one bank serialize the access
divergence	lanes of one warp taking different sides of a branch, so the warp runs both sides with lanes masked off
predication	the branchless form of an <code>if</code> : every lane runs the instruction and a per-lane flag decides whether the result is kept
<code>__syncthreads()</code>	the block-wide barrier: every thread of the block must arrive before any may continue; all threads must reach it
shuffle	a warp instruction (<code>__shfl_down_sync</code>) letting one lane read another lane's register directly
<code>atomicAdd</code>	an addition that is safe when many threads target the same address; serializes under heavy contention
grid-stride loop	a loop where each thread strides by the total thread count, letting a fixed grid handle any input size with coalesced access
constant memory	a 64 KB read-only region with a per-SM cache that broadcasts warp-uniform reads and can fold values into instructions as operands
halo (ghost cells)	the border of extra input a tile needs beyond its own outputs, shared with (and re-read by) neighboring blocks
arithmetic intensity	a kernel's useful floating-point operations divided by the bytes it must move; decides which ceiling applies
roofline / ridge	the two-ceiling performance model / the intensity (~20.3 here) where the memory and compute ceilings meet
effective bandwidth	ideal (must-move) bytes divided by time — a score against perfection rather than a report of actual traffic
<code>__restrict__</code>	the programmer's promise that a pointer's memory is accessed through that pointer alone, enabling the read-only load path

SASS	the GPU's real assembly language, viewable with <code>cuobjdump -sass</code> ; the ground truth of what the compiler produced
LDG / LDS / LDC / FFMA	assembly names for: a global load (suffix .CI marks the read-only path), a shared-memory load, a constant-memory load, and a fused multiply-add

14 What comes next

The roadmap has three kernels left, each chosen because it teaches a technique nothing so far has needed. The **histogram** introduces a new failure mode: thousands of threads all trying to *write* to the same few counters, which turns atomic operations from a footnote into the bottleneck; the technique that addresses it is privatization — giving each block its own private copy of the counters in shared memory and merging the copies at the end. **Matmul double-buffering** returns to chapter 5's unfinished business: the register-tiled kernel still reaches only 26% of the arithmetic peak, and the next step is to overlap the loading of the next tile with computation on the current one, so the arithmetic units never wait. And **sparse matrix-vector multiplication** (SpMV in CSR format) abandons the regular, predictable access patterns that every kernel so far has enjoyed, forcing the question of how to balance load when different rows contain wildly different amounts of work.

There is also one optional return trip: the 2D convolution ends chapter 11 spending three quarters of its time on shared-memory loads, and the known cure is the register-tiling idea from chapter 5 — several outputs per thread, so that each loaded value feeds several multiply-adds. Whether to take that trip is a judgment call about diminishing returns, which is itself a skill this project is meant to teach.

Interactive companions to these notes, each a self-contained page that opens in any browser:

`docs/register_tiled_matmul_viz.html`, `docs/bank_conflict_viz.html`,
`docs/memory_hierarchy_occupancy_viz.html`, `docs/blelloch_scan_tree_viz.html`, and
`docs/conv2d_halo_tiling_viz.html`.